

- 39** Tetanus is an infectious disease caused by a type of bacterium. This bacterium produces a protein that is a toxin which causes illness.

Scientists have produced a vaccine for tetanus which contains a harmless form of the toxin called a toxoid.

The toxoid is produced by mixing the toxin with the chemical formaldehyde. This chemical binds to the toxin making it harmless.

Which statement about the tetanus vaccine is correct?

- A** After vaccination, the toxoid in the vaccine will be ingested by neutrophils and displayed in their cell surface membranes.
- B** The tetanus vaccine containing the toxoid does **not** contain antigens because it does **not** contain any cells of the pathogen.
- C** The toxoid protein in the vaccine stimulates the production of antibodies which remain in the blood to provide long-term immunity.
- D** When formaldehyde binds to the toxin, it causes the toxin to completely change shape.